FINANCIAL MANAGEMENT

Time allowed- 3:30 hours
Total marks- 100

May - 2026

[N.B. - The figures in the margin indicate full marks. Questions must be answered in English. Examiner will take account of the quality of language and of the manner in which the answers are presented. Different parts, if any, of the same question must be answered in one place in order of sequence.]

Marks

1. Surma Pharmaceuticals PLC is planning to finance a new vaccine research facility costing BDT 2,000 million. The company is evaluating three distinct financing options to maintain its competitive edge and return on equity.

Current Financial Position:

- Shares Outstanding: 100 million shares.
- Current Market Price: BDT 160 per share.
- Existing Debt: BDT 4,000 million in 10% irredeemable bonds (at par).
- Current Earnings Before Interest and Tax (EBIT): BDT 1,800 million.
- Corporate Tax Rate: 22.5%.
- Equity Beta (β): 1.40 | Risk-Free Rate: 7.5% | Equity Risk Premium: 5.0%

Financing Options:

1. **Equity Issue:** Issue new shares at a 10% discount to the current market price.
2. **Debt Issue:** Issue 12% long-term debentures at par.
3. **Sukuk (Islamic Bond):** Issue a 5-year Shariah-compliant Sukuk with an expected profit rate of 11%, involving an asset-backed structure with real estate.

REQUIREMENTS:

a) Calculate the Weighted Average Cost of Capital (WACC) for the company under its current capital structure, using the Capital Asset Pricing Model (CAPM) to derive the cost of equity. 5

b) Calculate the Earnings Per Share (EPS) for both the Equity and Debt financing options, assuming EBIT increases by 15% following the investment. 4

c) Calculate the Financial Break-even Point — i.e., the EBIT level at which the EPS for the equity and debt options would be equal. Advise management based on the projected EBIT. 3

d) Evaluate the strategic benefits of choosing the Sukuk option over conventional debt in the Bangladesh context, considering liquidity, investor base, and Shariah-compliance. 3

2. Bengal Textiles PLC, a garment exporter in Dhaka, reported FY2025 revenues of BDT 4,500 million. The company exports to Europe, where buyers demand sustainability compliance. To meet export orders and EU audits, the company plans to expand factory capacity. Investment requirement is BDT 100 million. At the March 2026 board meeting, three financing options were presented:

Bank Loan: A 5-year facility at 12% interest. The loan requires interest-only payments for the first two years, followed by equal annual installments of principal and interest for the remaining three years. An application fee of 1% is payable upfront, and an insurance premium of 2% applies for the first six months. The agreement stipulates an early settlement penalty of 2% of outstanding principal, although early repayment is considered remote. The corporate tax rate is 30%, and all interest payments are deductible. Collateral and dividend restrictions apply.

Convertible Bond: A 10% coupon bond with a 5-year maturity, redeemable at par of BDT 100 unless converted. Conversion will occur if the share price exceeds BDT 120 at maturity. Market analysts project the share price will exceed BDT 124 at maturity. The company must buy shares from the market to honor the conversion call. An application fee of 0.5% is payable upfront. Coupon payments are deductible at the 30% corporate tax rate.

Right Share Issue: A rights offering to existing shareholders at a 20% discount, cost of equity 15%. Flotation costs 3%, application fee 0.5%, and underwriting commission 1%. Equity financing provides no tax shield, and issue expenses are not deductible.

The Chairman emphasized his long-standing role and warned that a rights issue or bond conversion could dilute his voting power, asking whether safeguards could protect his position. The Finance Director noted that debt would preserve ownership but cautioned that rising wages and volatile cotton prices could strain cash flows and make fixed interest obligations risky. The Independent Director

argued that equity would strengthen the balance sheet, reassure buyers and rating agencies, and pointed out that competitors in Vietnam and India had raised equity to signal resilience, suggesting Bengal Textiles may need to follow suit.

Requirements:

a) Calculate the annual cash outflows and the effective after-tax cost of financing under each option. 12

b) Analyze the Chairman's concern. If a rights issue is made, how can he retain control? 3

c) Evaluate the trade-offs between debt and equity in terms of cost, risk, and control, from at least two perspectives. 5

3. Delta Apparel Ltd. (DAL) is a Bangladesh-based exporter of high-quality garments supplying to European buyers. Due to increasing volatility in exchange and interest rates, the board is reviewing whether hedging policies should be implemented. One of DAL's key customers is Nordic Retail Group, based in Norway. DAL has secured a contract (DAL/EXP/26/01) to export garments. Production will start in October 2025, and delivery will be completed by September 2026. DAL will receive NOK 18 million on 30 September 2026. DAL currently holds surplus funds in BDT, which it plans to invest in March 2026 for six months.

Information regarding the Bangla and Norwegian currencies is given in the table below:

(1) Forecast Spot Exchange Rate (NOK/BDT) on 30 September 2026

Range	Probability
10.70 – 10.80	20%
10.80 – 10.90	30%
10.90 – 11.00	30%
11.00 – 11.10	20%

(2) Current Spot Rate
NOK/BDT = 10.80 – 10.95

(3) 12-Month Forward Rate
Forward margin: NOK at discount of 0.20 – 0.25

(4) Interest Rates (Annual)

Country	Borrowing	Deposit
Bangladesh	12%	10%
Norway	8%	6%

(5) Currency Options (12-month maturity)
DEL's bank offers a put option on NOK 18 million at an exercise price (NOK/BDT) of 10.85. A call option on NOK 18 million at an exercise price (NOK/BDT) of 10.75. The premium is Tk. 250,000

(6) Interest Rates (Annual)
DEL expects to invest Tk. 120 million in March 2026 for 6 months. It would like to investigate how it might hedge against adverse interest rate movements in Bangladesh. Current 6 month interest rate is 8%

(7) Available hedging instruments:
Forward Rate Agreement (FRA)
Interest rate futures (price = 92.00)
Interest rate options
Interest rate swaps

Requirements:

a) Assuming the current date is 30 September 2025, calculate the BDT amount receivable by DAL on 30 September 2026 if it uses: 6
   (i) the expected spot rate in 12 months' time
   (ii) a forward contract

b) Advise DAL's board as to the effectiveness of employing the following methods of hedging the company's exposure to interest rate risk on the proposed investment: 4
   (i) A Forward Rate Agreement (FRA)
   (ii) An interest rate future
   (iii) An interest rate option
   (iv) An interest rate swap

c) Calculate the BDT proceeds if DAL uses: 5
   (i) a money market hedge
   (ii) a currency option (best-case and worst-case outcome)

4. On 31 March 2026, Index Asset Management Ltd., a mid-sized institutional investor, holds a diversified portfolio of DSE 50 shares valued at BDT 39 million. The company intends to liquidate this portfolio on 30 June 2026 to fund construction of a new plant in Narayanganj.

The treasury report highlighted several market developments. The DSE 30 index rose 3% in March, driven by banking and telecom stocks. Trading in bank stock options was unusually active, with volumes nearly double the monthly average. Foreign investors preferred call options at higher strikes. July contracts showed wider bid ask spreads, reflecting lower liquidity and higher costs. Analysts noted institutional hedging demand was pushing up June premiums, while retail investors speculated heavily in short dated May options.

The treasury manager is concerned the DSE 50 index may decline, given uncertainty around monetary policy and possible regulatory changes in banking. The current DSE 50 index is 4,200 points. Each option contract is worth BDT 10 per index point, expiring at month end. The risk committee emphasized that financing the new plant requires a minimum of BDT 38 million in liquid funds by 30 June 2026. Any shortfall would require costly bridge financing from local banks at double digit interest rates. The committee asked the treasury manager to evaluate hedging strategies that balance downside protection with upside participation, while considering transaction costs and liquidity risks.

Option Premiums (BDT):

	Strike 4,100	Strike 4,200	Strike 4,300
May	Call 210 / Put 135	Call 150 / Put 160	Call 130 / Put 240
June	Call 245 / Put 170	Call 165 / Put 185	Call 155 / Put 265
July	Call 275 / Put 195	Call 185 / Put 205	Call 170 / Put 290

Requirements:

a) Demonstrate how June put options hedge the portfolio, calculate contracts and payoff, and briefly explain how protective puts guarantee a minimum value through the strike price and why proper sizing ensures effectiveness. 6

b) Compute portfolio outcome with and without hedging if sold for BDT 36 million at index 3,850, and if sold for BDT 42 million at index 4,350. 5

c) Assess whether the company should hedge fully or partially, considering the risk-return trade-off. Briefly explain how full hedging ensures funding certainty while partial hedging retains upside but leaves residual risk. 4

5. Green Power Bangladesh Ltd (GPBL) manufactures solar home systems and has a financial year ending 30 June. The company is considering a new investment project called 'Project Uttaran'. If the project is undertaken, GPBL will invest Tk. 32,000,000 in new machinery on 30 June 2026. The project life is 4 years (2027–2030).

The company has paid for market research which produced the following estimates for 'Project Uttaran':

Year to 30 June 2027 (all figures expressed in June 2026 prices)

Sales revenue	45,000,000
Variable cost	24,000,000
Fixed cost (excluding depreciation)	8,500,000
Increase in sales volume per year	12%
Inflation rates:	
Sales prices	7%
Variable cost	10%
Fixed cost	6%
Working capital (to be in place at the start of each year)	15% of sales
Residual value of capital equipment (in 2030 prices)	10,000,000
Tax and capital allowances:	
Tax rate	27.5%
Capital allownaces: at 20% reducing balance	
Other information:	
Nominal weighted average cost of capital	16%
Unless otherwise stated all cash flows occur at the end of the year	

Requirements:

a) Calculate the net present value of the project at 30 June 2026 and advise the company's directors whether they should proceed with it. 12

b) Calculate sensitivity of NPV to a 15% fall in sales volume (ignore working capital). 4

c) Discuss macroeconomic risks in Bangladesh affecting the project. 4

6. Delta Engineering Ltd. (DEL) had operated for forty years as a supplier of heavy machinery to government projects. Its revenues were stable, but the company's dependence on carbon intensive contracts was becoming a liability. Investors were questioning DEL's valuation, and regulators were tightening compliance requirements. The board concluded that without diversification, DEL's earnings multiple would continue to compress.

Renewable Power Technologies Ltd. (RPTL), a twelve-year-old solar energy company, was attracting attention. Its revenues were smaller, but its growth rate was high, supported by heavy research and development expenditure. RPTL's shares traded at a premium price earnings ratio, reflecting investor confidence in its ESG credentials and access to green financing.

The two boards began merger discussions. DEL sought entry into renewables to protect its cash flows and improve its cost of capital. RPTL wanted DEL's distribution network and government relationships to accelerate scale. After negotiation, they agreed on a 0.5 to 1 share swap, meaning each DEL share would be exchanged for half a share of RPTL. DEL shareholders argued the ratio undervalued their stable earnings base, while RPTL's management insisted its higher growth and ESG premium justified the terms.

Financial data framed the debate. DEL reported BDT 12,000 million revenue, BDT 1,200 million net profit, 200 million shares at BDT 60. RPTL reported BDT 8,000 million revenue, BDT 800 million net profit, 100 million shares at BDT 120. Post merger, 100 million new shares would be issued, raising total shares to 200 million. Synergies were estimated at BDT 400 million. Expansion would be financed by a BDT 5,000 million sustainability linked bond, with interest tied to carbon reduction performance. The coupon would be 6 percent if targets were met, 8 percent if missed.

DEL shareholders questioned fairness in the swap ratio and dilution. Some directors worried the merger could be perceived as greenwashing if ESG targets were not genuinely pursued. Employees raised concerns about the proposed Sustainability-linked Employee Share Ownership Plan (SLESOP), which promised additional shares if carbon emissions fell 30 percent in five years. While the plan aligned incentives with ESG goals, staff feared the targets might be unrealistic and create pressure. The merger promised synergies, improved access to capital, and are rerating of valuation multiples, but fairness, transparency, and sustainability credibility remained contested.

Requirements:

a) Using the financial data assess whether the 0.5 to 1 swap ratio is financially fair to DEL. 8

b) Identify and evaluate the ethical issues in valuation, negotiation, and sustainability claims, considering shareholder dilution and reputational risks. 3

c) Assess the SLESOP design and recommend how the merged entity should balance ESG commitments with shareholder value. 4

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